

Football Project – Research Questions Proposal (Final Version)

Purpose

The following research questions have been specifically designed around the structure and available variables of our football dataset. They maximize the analytical value of the available data while maintaining a coherent project narrative. The questions intentionally share several variables, allowing us to perform a single data cleaning process and build a consistent EDA and SQL analysis throughout the project.

Top 4 Recommended Research Questions

1. How accurately do Expected Goals (xG) predict the final match result?

Why? Expected Goals (xG) is one of the most important metrics in modern football analytics.

Main Variables: Home xG, Away xG, Home Goals, Away Goals, League, Season.

Possible Analysis: Correlation between xG and goals, comparison across leagues and seasons.

2. Does recent team performance influence match outcomes?

Why? The dataset contains extensive information about recent team performance.

Main Variables: Previous Points, Previous Goals Scored, Previous Goals Conceded, Previous xG, Match Result, League.

Possible Analysis: Team form, winning probability and performance trends.

3. Do teams with more current points consistently achieve better match results?

Why? Evaluate whether the number of points accumulated before a match predicts future performance.

Main Variables: Current Points, Goals, xG, Match Result, League, Season.

Possible Analysis: Current points vs wins, goals and Expected Goals.

4. How do team player ratings and team formations influence match performance?

Why? Uses team-level player ratings together with tactical formations.

Main Variables: Team Player Ratings (Defense, Midfield, Attack), Formation, Goals, xG.

Possible Analysis: Ratings vs goals, ratings vs xG, best formations and league comparisons.

Additional Research Questions

5. Which league has the highest average Expected Goals (xG) per match?

6. Which league produces the highest average number of goals per match?

7. Is there a relationship between Expected Goals (xG) and actual goals scored?

8. Which league is the most competitive based on goal difference and match results?

9. Does home advantage vary across Europe's Top Five Leagues?

10. How does team form affect Expected Goals (xG) in future matches?

11. Which formations are associated with the highest attacking performance?

12. Do teams with higher player ratings consistently outperform lower-rated teams?

Variables Shared Across the Recommended Questions

Variable	Q1	Q2	Q3	Q4
League	✓	✓	✓	✓
Season	✓	✓	✓	✓
Goals	✓	✓	✓	✓
Expected Goals (xG)	✓	✓	✓	✓
Current Points			✓	
Previous Performance		✓		
Team Player Ratings				✓
Formation				✓

Recommended Project Title

Performance Analysis of Europe's Top Five Football Leagues: Understanding the Factors Behind Match Outcomes

Final Recommendation

The four recommended research questions provide a coherent project structure, efficiently use the available dataset, share key variables, require only one data cleaning process, and support a comprehensive EDA and SQL analysis.